Quiz 11

1. Consider the Lagrangian for a charge in an electric field given by:

$$\mathcal{L} = \frac{1}{2}m\dot{\vec{r}}^2 - q(V - \dot{\vec{r}} \cdot \vec{A}), \tag{1}$$

where $\vec{A}$ and V are related to the electric field by:

$$\vec{E} = -\vec{\nabla}V - \frac{\partial \vec{A}}{\partial t} \tag{2}$$

(a) Write down the Lagrangian assuming the charge can only move in the $\hat{x}$ direction (so that $\vec{r} = x\hat{x}$, $\dot{\vec{r}} = \dot{x}\hat{x}$.) Use this assumption for the rest of the problem

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - q\left[V - (\dot{x}\hat{x}) \cdot \vec{A}\right] = \frac{1}{2}m\dot{x}^2 - q(V - \dot{x}A_x)$$

(b) Find $\frac{\partial \mathcal{L}}{\partial x}$.

$$\frac{\partial \mathcal{L}}{\partial x} = -q\frac{\partial V}{\partial x} + q\dot{x}\frac{\partial A_x}{\partial x}$$

(c) Find $\frac{\partial \mathcal{L}}{\partial \dot{x}}$.

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} + qA_x$$

(d) Find $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}$. Note that $\vec{A}(x, y, z, t)$ and use the chain rule when taking the time derivative.

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\ddot{x} - q\frac{dA_x}{dt} = m\ddot{x} + q\left[\frac{\partial A_x}{\partial x}\frac{dx}{dt} + \frac{\partial A_x}{\partial y}\frac{dy}{dt} + \frac{\partial A_x}{\partial z}\frac{dz}{dt} + \frac{\partial A_x}{\partial t}\right]$$

$$= m\ddot{x} + q\frac{\partial A_x}{\partial x}\dot{x} + q\cancel{\frac{\partial A_x}{\partial y}}^{\,0}\dot{y} + q\cancel{\frac{\partial A_x}{\partial z}}^{\,0}\dot{z} + q\frac{\partial A_x}{\partial t}$$

$$= \boxed{m\ddot{x} + q\frac{\partial A_x}{\partial x}\dot{x} + q\frac{\partial A_x}{\partial t}}$$

(e) Find the equation of motion for a charge in an electric field.

$$-q\frac{\partial V}{\partial x} + \cancel{q\dot{x}\frac{\partial A_x}{\partial x}} = m\ddot{x} + \cancel{q\frac{\partial A_x}{\partial x}\dot{x}} + q\frac{\partial A_x}{\partial t}$$

$$\boxed{-q\frac{\partial V}{\partial x} = m\ddot{x} + q\frac{\partial A_x}{\partial t}}$$

(f) Show that $m\ddot{x} = qE_x$

$$E_x = -\frac{\partial V}{\partial x} - \frac{\partial A_x}{\partial t}$$

$$m\ddot{x} = -q\frac{\partial V}{\partial x} - q\frac{\partial A_x}{\partial t}$$

$$m\ddot{x} = -qE_x \quad \checkmark$$